

did you know?

More than 2,000 years ago, Greek thinker Aristotle recommended that people study nature and conduct experiments to test the accuracy of ideas.

A kite helped Benjamin Franklin learn about lightning and electricity.

Benjamin Franklin (1706–1790)

American scientist Benjamin Franklin experimented with lightning and electricity. His work in the 1700s laid the foundations for today's electrical world.



Franklin risked his life flying a kite—he could have been struck by lightning.

Louis Pasteur (1822–1895)

Pasteur is known for discovering pasteurization—a process that uses heat to destroy bacteria in food, particularly milk. He also found that some diseases are caused by germs and encouraged hospitals to be very clean to stop the spread of germs.



Super inventions!

Inventions and discoveries have changed the course of our history.



The first known **wheel** was used in Mesopotamia around 3500 BCE.



Paper was invented in China around 105 CE, but kept secret for many years.



The magnetic **compass** was first used by the Chinese. It was invented around 247 BCE.



The **parachute** was first tested in 1617 by Faust Vrancic, centuries after da Vinci made his drawings.



The **steam engine** was invented in 1804. The earliest successful model reached 30 mph (48 kph).



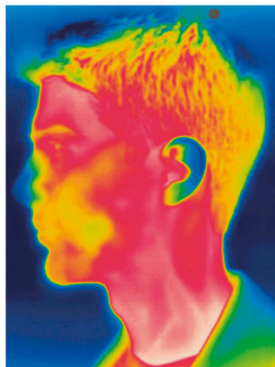
The **color photo** was first produced by physicist James Maxwell in 1861.

1700

1800

William Herschel (1738–1822)

Herschel is well known for his work in astronomy (he was the first to identify the planet Uranus). He also discovered infrared radiation—this technology is used today for wireless communications, night vision, weather forecasting, and astronomy.



Wilhelm Conrad Röntgen (1845–1923)

Röntgen discovered electromagnetic rays—today known as X-rays—on November 8, 1895. This important discovery earned him the first Nobel Prize for Physics in 1901.

X-rays allow doctors to look inside the human body.

